

Capital.com Public API - Markets, Bid/Ask & Price History

Part 2 for CapitalDemoBroker / Codex

Developer reference prepared from the official Capital.com Public API 1.0.0 documentation. Technical endpoint names, field names, enums and example structures are retained where needed for implementation; descriptive text is condensed.

1. Market identification: epic

Capital.com uses **epic** as the market/instrument identifier. A human search term can be resolved with GET `/api/v1/markets?searchTerm=...`. For the bot, do not assume that the display symbol is the broker epic: resolve and persist the exact epic.

```
GET /api/v1/markets?searchTerm=gold
```

Headers:

CST: REDACTED_CST

X-SECURITY-TOKEN: REDACTED_SECURITY_TOKEN

DEMO:

<https://demo-api-capital.backend-capital.com/api/v1/markets?searchTerm=gold>

2. GET /markets - searchable market list

Parameter	Type	Notes
searchTerm	string	Search term; takes priority if epics is also supplied
epics	string	Comma-separated epics; maximum 50

Market-list responses expose fields useful for discovery and live snapshots, including epic, symbol, instrumentName, instrumentType, lotSize, bid, offer, marketStatus, streamingPricesAvailable, pipPosition, tickSize, updateTimeUTC, high and low.

Representative structure:

```
{
  "markets": [{
    "epic": "SILVER",
    "symbol": "Silver",
    "instrumentType": "COMMODITIES",
    "instrumentName": "Silver",
    "lotSize": 1,
    "bid": 24.366,
    "offer": 24.386,
    "marketStatus": "TRADEABLE",
    "streamingPricesAvailable": true,
    "pipPosition": 2,
    "tickSize": 0.001,
    "updateTimeUTC": "2022-04-06T13:17:38.477"
  }]
}
```

3. GET /markets/{epic} - full instrument constraints

```
GET /api/v1/markets/{epic}
```

DEMO:

<https://demo-api-capital.backend-capital.com/api/v1/markets/{epic}>

Headers:

CST: REDACTED_CST

X-SECURITY-TOKEN: REDACTED_SECURITY_TOKEN

The response is split into **instrument**, **dealingRules** and **snapshot**. These sections should be treated as the authoritative broker constraints before submitting size, SL, TP or working-order prices.

Section / field	Purpose for CapitalDemoBroker
instrument.epic	Exact broker market ID
instrument.symbol / name / type	Identity and display metadata
instrument.lotSize	Contract/lot metadata
instrument.currency	Instrument currency
instrument.marginFactor / marginFactorUnit	Margin requirement
instrument.guaranteedStopAllowed	Whether guaranteed stops are supported
instrument.streamingPricesAvailable	Whether streaming quotes are available
instrument.openingHours + zone	Trading sessions; documented example uses UTC
instrument.overnightFee	Long/short overnight rates and charge timing
dealingRules.minStepDistance	Minimum price-step constraint
dealingRules.minDealSize / maxDealSize	Allowed deal-size range
dealingRules.minSizeIncrement	Position-size increment
dealingRules.minGuaranteedStopDistance	Guaranteed-stop minimum
dealingRules.minStopOrProfitDistance	Minimum SL/TP distance
dealingRules.maxStopOrProfitDistance	Maximum SL/TP distance
snapshot.marketStatus	Current market state
snapshot.bid / offer	Current REST Bid and Ask
snapshot.decimalPlacesFactor	Price decimal precision metadata
snapshot.scalingFactor	Scaling metadata

Documented example fragments:

```
"dealingRules": {
  "minStepDistance": {"unit": "POINTS", "value": 0.001},
  "minDealSize": {"unit": "POINTS", "value": 1},
  "maxDealSize": {"unit": "POINTS", "value": 100000},
  "minSizeIncrement": {"unit": "POINTS", "value": 0.1},
  "minGuaranteedStopDistance": {"unit": "PERCENTAGE", "value": 1},
  "minStopOrProfitDistance": {"unit": "PERCENTAGE", "value": 0.01},
  "maxStopOrProfitDistance": {"unit": "PERCENTAGE", "value": 60}
},
"snapshot": {
  "marketStatus": "TRADEABLE",
  "bid": 22.041,
  "offer": 22.061,
  "decimalPlacesFactor": 3,
  "scalingFactor": 1
}
```

4. Bid / Ask naming

REST market data uses **bid** and **offer** in market snapshots. Historical candle objects use **bid** and **ask**. Therefore the adapter should normalize Capital.com's REST offer to the gateway's Ask field.

Spread can be derived at a snapshot as Ask - Bid, but the API documentation does not define that subtraction as a separate authoritative 'spread' field in these responses.

5. Historical prices / 1-minute candles

GET /api/v1/prices/{epic}

Query parameters:

```
resolution = MINUTE | MINUTE_5 | MINUTE_15 | MINUTE_30 |
              HOUR | HOUR_4 | DAY | WEEK
max         = default 10, maximum 1000
from        = YYYY-MM-DDTHH:MM:SS
```

to = YYYY-MM-DDTHH:MM:SS

DEMO:

<https://demo-api-capital.backend-capital.com/api/v1/prices/{epic}>

Example:

GET /api/v1/prices/GOLD?resolution=MINUTE&max=10

The official one-minute resolution token is **MINUTE**. Date filtering is based on **snapshotTimeUTC**. If no query parameters are supplied, the endpoint defaults to minute prices for the last 10 minutes.

6. Candle structure

```
{
  "prices": [{
    "snapshotTime": "2022-04-06T15:18:00",
    "snapshotTimeUTC": "2022-04-06T13:18:00",
    "openPrice": { "bid": 24.356, "ask": 24.376 },
    "closePrice": { "bid": 24.378, "ask": 24.398 },
    "highPrice": { "bid": 24.378, "ask": 24.398 },
    "lowPrice": { "bid": 24.355, "ask": 24.375 },
    "lastTradedVolume": 187
  }]
}
```

Capital.com therefore supplies separate Bid and Ask OHLC values in the historical-price response. The public REST section shown here does not supply a separate Mid OHLC field.

7. Entry Filter implication

For the 1-minute Entry Filter, the API gives enough data to build Bid candles, Ask candles, or a locally derived midpoint candle. However, the documentation does not state that the strategy must use Bid, Ask or Mid. That choice is a strategy-level decision and should not be silently invented inside CapitalDemoBroker.

Likewise, the REST contract provides timestamped bars but the cited section does not explicitly label a returned bar as 'closed' versus 'currently forming'. The implementation should determine closed/current status from UTC timestamps and current time only after defining the intended boundary rule in the project contract.

8. WebSocket market data

Capital.com also supports WebSocket real-time market data. The documentation limits subscriptions to a maximum of 40 instruments at a time. OHLC subscriptions use destination **OHLCMarketData.subscribe**; the default resolution is MINUTE and supported resolutions match the REST set.

OHLC subscription payload:

```
{
  "destination": "OHLCMarketData.subscribe",
  "correlationId": "3",
  "cst": "REDACTED_CST",
  "securityToken": "REDACTED_SECURITY_TOKEN",
  "payload": {
    "epics": ["GOLD"],
    "resolutions": ["MINUTE"],
    "type": "classic"
  }
}
```

OHLC event fields:

```
resolution
epic
type
priceType
t (timestamp)
h high
l low
o open
c close
```

A documented OHLC streaming event includes **priceType** (the example is 'bid'), so consumers must not assume that every OHLC event is a midpoint candle.

9. Trading-hours / market-status safeguards

Single-market details include openingHours with an explicit zone and snapshot.marketStatus. The documented examples include TRADEABLE and CLOSED states in market responses. CapitalDemoBroker should fetch/refresh market state and reject new-cycle trading when the market is not confirmed tradeable.

10. What is confirmed vs. still unresolved

Confirmed by API docs	Still must not be guessed
GOLD-like names can be resolved through GET /markets?searchTerm=...	Which returned epic is the intended GOLD if multiple matches exist
Current REST quote has bid + offer	Strategy candle price basis: Bid vs Ask vs derived Mid
1-minute resolution token is MINUTE	Exact project rule for identifying current vs fully closed candle
Historical OHLC has separate bid + ask	Whether Entry Filter should use broker historical REST or streaming OHLC
minDealSize, minSizeIncrement and stop/profit distance constraints are exposed	Exposing direction for every order field unless explicitly established elsewhere
openingHours and marketStatus are exposed	Holiday exceptions beyond what the current market metadata/status reports

11. Recommended CapitalDemoBroker mapping

- resolve_market(query) -> GET /markets?searchTerm=...
- market_details(epic) -> GET /markets/{epic}
- quote(epic) -> snapshot.bid + snapshot.offer (normalize offer -> ask)
- candles_1m(epic, ...) -> GET /prices/{epic}?resolution=MINUTE
- validate_size() -> minDealSize, maxDealSize, minSizeIncrement
- validate_protection_levels() -> min/max stop-or-profit distances + price precision metadata
- is_tradeable() -> snapshot.marketStatus plus opening-hours awareness

12. Source

Official source: Capital.com Public API 1.0.0, Markets, Prices and WebSocket market-data sections, accessed 10 Aug 2026: <https://open-api.capital.com/>